



Fig. 15: Put the copper-clad board in the FeCl_3 solution (left) and constantly tilt the container from side to side without spilling its contents (right)

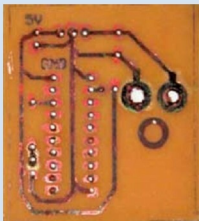


Fig. 16: The board just taken out of FeCl_3 solution

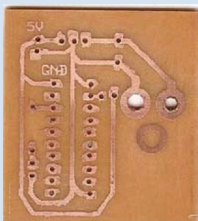


Fig. 17: The PCB is ready after removing carbon using a scrubber

appears normal on the copper side of the PCB.

3. Always print from a laser printer. Printout from inkjet or any other printer will not work as it may not be sharp enough.

4. You can print on both top and bottom parts of the board, but here we will print text only on the top copper layer.

5. There are standards for using traces of different thicknesses for different purposes:

- (a) 0.6mm (0.025-inch) trace for single tracks
- (b) 1.3mm (0.050-inch) trace for power and ground tracks
- (c) 0.2mm to 0.4mm (0.010- to 0.015-inch) traces for going between IC and component pads

6. Tracking from large to small and then back to large is known as 'necking-in' or 'necking down' (see Fig. 4).

7. You can also let rest of the space

remain covered by copper, leaving clearance beside each line, so our PCB will look as shown in Fig. 5. We will not use it as this type of filled circuit is best for industrial-level machine etching.

8. The standards for clearances for electrical conductors are listed here in the table (on previous page).

9. For thin traces (<0.6mm traces), it is good to add 'chamfer' to 'T' junction, thus eliminating 90° angles (see Fig. 6).

Printing the PCB layout

Go to File→Print. A window as shown in Fig. 7 will open. Select 'Layers to Print'. Keep 'Enlarge to Fit Page' option unchecked as it will not give the exact layout of the components. Print the PCB layout from a laser printer. If you don't have a laser printer at home, convert the file into a PDF file (using PDF995 software) and then print it elsewhere.

The printout of the PCB layout looks as shown in Fig. 8.

Etching

Cut the copper-clad board to a size matching the size of the PCB design printout. You can also use a glass epoxy board but it's costlier than a copper-clad board. Fig. 9 shows a full-size copper-clad board.

Put the paper printout on the board with the printed side facing the copper side. Affix the paper to the board using cello tape so that the paper does not move while ironing (Fig. 10).

Now take your household electric iron and set its temperature to the maximum (Fig. 11). Press the hot iron on the paper and carefully move it across the paper for about four minutes. While doing so, check for impressions. Continue ironing until the complete impression of the circuit comes on the copper side (Fig. 12).

Complete the blanks, if any, with a good permanent marker. If any line is not dark enough, redraw it on the board using the permanent marker. Wash the board in normal tap water.

Drill IC holes using a 1mm hand PCB drill as shown in Fig. 14. Redraw the lines using the permanent marker if they have been defaced by the drill.

Now mix some FeCl_3 (ferric chloride) powder in hot water. The reaction is vigorous, so take safety precaution. Put the copper-clad board in the solution and constantly tilt the container from side to side without spilling its contents (Fig. 15). This is done to speed up the reaction. It takes five to six minutes to wash away all the excess copper. In between, keep checking the board. Ensure that the marker or carbon of the impression does not wash away.

Take the board out (Fig. 16) and wash it under tap water to remove the permanent marker ink. Remove carbon by using nail polish remover. Use a scrubber to gently scrub the copper surface (Fig. 17) till it shines. Your PCB is ready! ●

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